

# Ship Rock

*A twin-peaked volcanic plug with spectacular radiating dikes, which stands high above the deserts of New Mexico*



The most famous volcanic plug in the American Southwest, Ship Rock towers 1,585 ft (483 m) above a high-desert plain in the Navajo Nation—a Native American territory in Arizona and New Mexico. Also called a volcanic neck, this type of feature originates when a mass of erosion-resistant rock forms inside a volcano after it has stopped erupting (see panel, right).

## Rock ribs

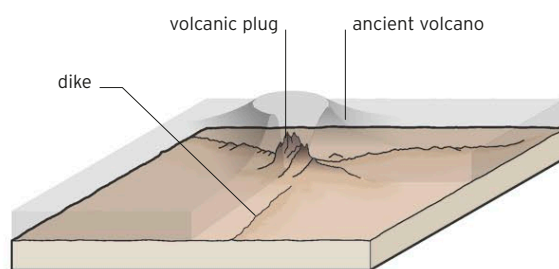
Regarded as a sacred monument by the Navajo people, Ship Rock's Navajo name is *Tsé Bit'a'í*, which means "winged rock." This probably alludes to the presence of prominent walls of volcanic rock, called dikes, that radiate from the monolith. The three most prominent of these rock ribs run at roughly 120 degrees to each other. Seen from certain perspectives, the whole monument can have a birdlike appearance.

## ▽ NAVAJO MONUMENT

Ship Rock and three of its dikes can be seen here. According to a Navajo legend, birdlike monsters once nested on the rock's high peaks, feasting on human flesh.

## HOW SHIP ROCK FORMED

Ship Rock is composed of a hard igneous rock called lamprophyre that solidified in the throat of an ancient volcano about 30 million years ago. The accompanying radiating dikes formed around the same time. The rest of the volcano, which consisted of softer rock, has since eroded away.



## △ KNIFE EDGE

Parts of Ship Rock's dikes have a knifelike form and tower at a height of 65 ft (20 m) above the high-desert plain.

The longest of Ship Rock's dikes stretches for nearly 2 miles (3 km)

